

Yasong Fan

Independent Researcher

arXiv: 2604.07716 github.com/YasongFan/FDM

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The Editors

Nature Machine Intelligence

Dear Editors,

Most linear recurrent sequence models compress all historical context into a single fixed-size state, coupling global information propagation and selective token retrieval in a way that makes both functions compete during training. This paper introduces FDM (Fan Duality Model), which separates these two functions into a wave channel and a cache channel, and shows that their joint training produces a gradient sink that leaves the cache severely undertrained. A phase-aware staged training strategy resolves this interference and substantially improves performance on both language modelling and genomic sequence tasks.

The key results are as follows. On WikiText-103, FDM reaches perplexity 33.75 under phase-aware training, below a similarly sized Transformer reference under the present training setup. Runtime memory saturates beyond moderate context lengths, in contrast to the linear growth of Transformer KV-cache requirements. Pretrained on 2.94 Gbp of human reference genome, FDM shows clear gains over HyenaDNA on promoter and enhancer recognition across three random seeds, approaches CADD on variant effect prediction after fine-tuning, and is consistent with partial cross-species transfer to *Drosophila* regulatory elements. Mutual information analysis provides empirical support for functional specialisation between the two channels.

We believe this work will be of interest to *Nature Machine Intelligence* readers working on sequence architecture design, efficient biological sequence modelling, and the mechanistic analysis of learned functional organisation in neural networks. The combination of an architectural contribution, a training-dynamics finding, and evaluation across two sequence domains fits the journal's scope for work that bridges machine learning methods and biological application.

The manuscript is original, has not been submitted elsewhere, and the author approves this submission. Code and pretrained weights are available at github.com/YasongFan/FDM.

Sincerely,

Yasong Fan